# Beomgu Kang, Ph.D.

**Name** Beomgu Kang (강범구) **Current Position** Postdoctoral Researcher, Korea University
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## Research Profile

My research focuses on reliable and resource-efficient AI, spanning physics-informed optimization, active sensing, and LLM inference. I develop methods for optimizing computational and measurement resources under uncertainty, with applications to language models, inverse problems, and medical imaging.

## Education

**Korea Advanced Institute of Science and Technology (KAIST)** Daejeon, South Korea
*Ph.D. in Electrical Engineering* *Aug. 2020 - Feb. 2024*
- **Advisor:** HyunWook Park
- **Thesis:** Learning-based Quantification of Magnetization Transfer Contrast (MTC) and Chemical Exchange Saturation Transfer (CEST) using MR Fingerprinting

*M.S. in Electrical Engineering* *Mar. 2019 - Aug. 2020*
- **Advisor:** HyunWook Park
- **Thesis:** Unsupervised Learning for Magnetization Transfer Contrast MR Fingerprinting (MTC-MRF) and Chemical Exchange Saturation Transfer (CEST) MRI

*B.S. in Electrical Engineering* (*Double Major: Bio and Brain Engineering*) *Mar. 2013 - Feb. 2019*

## Academic & Research Experience

**Korea University, Department of Artificial Intelligence** Nov. 2025 – present
*Postdoctoral Researcher* *Seoul, South Korea*
- **Advisor:** Hyunseok Seo

**Korea Institute of Science and Technology (KIST), Bionics Research Center** Mar. 2024 – Oct. 2025
*Postdoctoral Researcher* *Seoul, South Korea*
- **Advisor:** Hyunseok Seo

**Johns Hopkins University School of Medicine, Department of Radiology** Dec. 2022 – Jan. 2023
*Visiting Scholar* *Baltimore, MD*
- **Advisor:** Hye-Young Heo

## Awards & Honors

**ISMRM Junior Fellow** 2025
**Editor's Monthly Pick, Magnetic Resonance in Medicine** 2023
**Editor's Monthly Pick, Magnetic Resonance in Medicine** 2021
**ISMRM Summa Cum Laude Merit Award** 2021

## Publications

### Journal Articles

**[J1]** Sultan Z. Mahmud, **Beomgu Kang**, Kevin Ju, Ji Eun Park, Doris D.M. Lin, Ouri Cohen, Christian T. Farrar, Or Perlman, and Hye-Young Heo. "Unleashing the potential of saturation-transfer MR fingerprinting (ST-MRF) in neuro-oncology: quantitative molecular CEST and MTC mapping." *Magnetic Resonance Materials in Physics, Biology and Medicine*, 2026.

**[J2]** **Beomgu Kang**, Munendra Singh, Hyunseok Seo, HyunWook Park, Hye-Young Heo. "Physics-informed optimization of saturation-transfer MRI protocols using non-differentiable Bloch models." *Physics in Medicine & Biology*, 2026.

**[J3]** Munendra Singh, **Beomgu Kang**, Sultan Z. Mahmud, Peter van Zijl, Jinyuan Zhou, Hye-Young Heo. "Saturation transfer MR fingerprinting (ST-MRF) for MTC and CEST quantification." *Magnetic Resonance in Medicine*, 2025.

**[J4]** **Beomgu Kang**, Wonil Lee, Hyunseok Seo, Hye-Young Heo, HyunWook Park. "Self-supervised learning for denoising of multidimensional MRI data." *Magnetic Resonance in Medicine*, 2024.

**[J5]** Jeewon Kim, Wonil Lee, **Beomgu Kang**, Hyunseok Seo, HyunWook Park. "A noise robust image reconstruction using slice aware cycle interpolator network for parallel imaging in MRI." *Medical Physics*, 2024.

**[J6]** **Beomgu Kang**, Munendra Singh, HyunWook Park, Hye-Young Heo. "Only-train-once MR Fingerprinting for B0 and B1 inhomogeneity correction in quantitative magnetization-transfer contrast." *Magnetic Resonance in Medicine*, 2023. **[Editor's Monthly Pick] [Top 10% Most-Viewed Paper in 2023]**

**[J7]** **Beomgu Kang**, Byungjai Kim, HyunWook Park, Hye-Young Heo. "Learning-based optimization of acquisition schedule for magnetization transfer contrast MR fingerprinting." *NMR in Biomedicine*, 2022.

**[J8]** **Beomgu Kang**, Byungjai Kim, Michael Schär, HyunWook Park, Hye-Young Heo. "Unsupervised learning for magnetization transfer contrast MR fingerprinting: Application to CEST and nuclear Overhauser enhancement imaging." *Magnetic Resonance in Medicine*, 2021. **[Editor's Monthly Pick] [Top Cited Article 2021–2022]**

## AI Conference Papers

**[A1]** Hojoon Kim, **Beomgu Kang**, Hyunseok Seo. "Not All Instances Should Contribute Equally: Sparse Support Grounding via Unbalanced Transport for Heterogeneous MIL." *Advances in Neural Information Processing Systems* (*NeurIPS*), 2026.

**[A2]** **Beomgu Kang**, Hyunseok Seo. "Prior-aware and Context-guided Group Sampling for Active Probabilistic Subsampling." *International Conference on Learning Representations* (*ICLR*), 2026.

**[A3]** **Beomgu Kang**, Hye-Young Heo, HyunWook Park. "Only-Train-Once MR Fingerprinting for Magnetization Transfer Contrast Quantification." *Medical Image Computing and Computer Assisted Intervention* (*MICCAI*), 2022. **[Early Accept: Top 13%]**

**[A4]** Yongjin Lee, Seokjun Park, **Beomgu Kang**, and HyunWook Park. "Multi-modal Characteristic Guided Depth Completion Network." *Proceedings of the Asian Conference on Computer Vision* (*ACCV*), 2022, pp. 4598–4612.

## Conference Proceedings and Workshops

**[C1]** **Beomgu Kang**, Hye-Young Heo, and Hyunseok Seo. "Taming Gradient Perturbations: Probabilistic Reparameterization for Robust Physics-Informed MRI Protocol Optimization." *ICML AI for Physics Workshop*, 2026.

**[C2]** **Beomgu Kang** and Hyunseok Seo. "Deep Probabilistic Multishell Undersampling for Diffusion Spectrum Imaging." *Proc. Intl. Soc. Mag. Reson. Med. 34*, 2026.

**[C3]** **Beomgu Kang**, Munendra Singh, Hyunseok Seo, Hye-Young Heo, and HyunWook Park. "Optimization of acquisition schedules and correction of B0 and B1 field inhomogeneities in saturation transfer MR fingerprinting (ST-MRF)." *Proc. Intl. Soc. Mag. Reson. Med. 33*, 2025.

**[C4]** Jeewon Kim, **Beomgu Kang**, HyunWook Park, and Hyunseok Seo. "Self-supervised deep learning model for denoising and distortion correction in accelerated echo planar imaging." *Proc. Intl. Soc. Mag. Reson. Med. 33*, 2025.

**[C5]** **Beomgu Kang**, Munendra Singh, Hye-Young Heo, and HyunWook Park. "Only-Train-Once MR Fingerprinting for B0 and B1 Inhomogeneity Correction in Quantitative Magnetization Transfer Contrast." *Proc. Intl. Soc. Mag. Reson. Med. 31*, 2023. **[Oral]**

**[C6]** Namho Jeong, **Beomgu Kang**, and HyunWook Park. "Brain tumor segmentation using out-of-distribution feature from normal MR images." *Proc. Intl. Soc. Mag. Reson. Med. 31*, 2023.

**[C7]** Seonghyuk Kim, Wonil Lee, Namho Jeong, Jeewon Kim, Jongyeon Lee, **Beomgu Kang**, and HyunWook Park. "A noise-robust accelerated MRI reconstruction using CycleGAN." *Proc. Intl. Soc. Mag. Reson. Med. 31*, 2023. **[Power Pitch]**

**[C8]** Jeewon Kim, Wonil Lee, **Beomgu Kang**, Seohee So, and HyunWook Park. "A noise robust image reconstruction deep neural network with cycle interpolation." *Proc. Intl. Soc. Mag. Reson. Med. 31*, 2023.

**[C9]** **Beomgu Kang**, Hye-Young Heo, and HyunWook Park. "Learning-based prediction of encoding capability for acquisition schedule of Magnetization Transfer Contrast MR fingerprinting." *Proc. Intl. Soc. Mag. Reson. Med. 30*, 2022.

**[C10]** **Beomgu Kang**, Byungjai Kim, Hye-Young Heo, and HyunWook Park. "Learning-based Optimization of Acquisition Schedule for Magnetization Transfer Contrast MR Fingerprinting." *Proc. Intl. Soc. Mag. Reson. Med. 29*, 2021. **[Oral] [Summa Cum Laude]**

**[C11]** **Beomgu Kang**, Byungjai Kim, Michael Schär, HyunWook Park, and Hye-Young Heo. "Unsupervised Deep Learning-based Magnetization Transfer Contrast (MTC) MR Fingerprinting and CEST MRI." *Proc. Intl. Soc. Mag. Reson. Med. 28*, 2020. **[Power Pitch]**

## Academic Service

**Conference Reviewing:** AAAI(2026-2027); ICML (2026 Silver Reviewer); NeurIPS(2026); ISMRM(2024-)

**Journal Reviewing:** Magnetic Resonance in Medicine (MRM); Transactions on Machine Learning Research (TMLR)